Risks Assessment

In this section, key risks that are both associated with the development of the ERPS and in scope with Tema 8's involvement with the project are identified and analysed. Each risk is appraised and ranked by the intersection of their impact on the project's development, and the likelihood of occurrence. Additionally, each risk identified is accompanied by a mitigation strategy.

Risk Matrix:

Overall threat level Key:

Green: Low	Orange: Medium	Red: High

Identified Risks

Rank	Risk	Impact	Likelihood	Mitigation Strategy
1	Inadequate risk management: Poor risk identification and analysis. No mitigation strategies are considered.	Major	Possible	Develop a risk register, with mitigation strategies included. Risks mentioned should include those found in research and from possible past risk registers of similar projects. Review and update the register frequently.
2	Fails to integrate with existing systems and structures.	Major	Possible	Conduct early analysis on all integration points, which will inform the system requirements. Conduct integration testing before final deployment.
3	Poor or substandard design	Major	Possible	Construct a detailed quality assurance program that must include testing done by end users.
4	Sudden power outages: Loss of power means the system will shut down. If this is sudden, it's unexpected, meaning any manual saves might not have been completed, and data may be lost.	Major	Possible	Include autosave functionality in the system.

5	Poor or non-existent internet connectivity: if the system is configured to require an internet connection, the lack thereof will render it unusable.	Major	Possible	Design the system with offline capabilities and develop performance while accounting for various network connectivity scenarios.
6	Poor data quality (i.e. incomplete records, inconsistent data entry and storage, errors with manual inputs)	Major	Possible	Ensure data verifications and validation controls.
7	Weak role-based access control: What should be locked behind certain ranks or responsibilities are not configured to express that.	Major	Possible	Define all role-specific workflows and set user access and authorisation to reflect that documentation.
8	Gaps in the system's privacy assurance during data collection, analysis, and storage.	Major	Possible	Mandate the completion of a Privacy Impact Assessment (PIA) before system deployment.
9	Unplanned team member unavailability: Team members are briefly yet suddenly unavailable, meaning they cannot compete their allocated tasks or contribute to the project development.	Moderate	Highly Likely	Document all assigned tasks and current contributions to streamline emergency reallocation of tasks. All current knowledge should also be documented to prevent misinformation. Each task should be assigned a backup executor, and thus, staff should be multi-skilled and up to date.
10	Loss of documentation: The only copy of documents (such as scope agreement, team charter, task allocation, and contribution tracing worksheets) are lost, stolen, or destroyed.	Major	Highly Likely	Store documentation digitally in a version-controlled repository. Conduct regular audits of the repository.
11	Scope Creep: additional features and requirements are continuously added without regard for resource, budget, or time constraints.	Moderate	Highly Likely	Define and document the scope through a scope statement, work breakdown structure, and product breakdown structure. Implement a formalised change control process that requires sign-off by the project sponsor.

12	Poor requirement gathering: Requirements gathered are incomplete and incorrect.	Moderate	Highly Likely	Have meetings with the stakeholders, document requirements and obtain final sign-off from stakeholders before project development can begin.
13	Poor team management: the team is in constant conflict which isn't resolved. Additionally, the team has limited cohesion and direction.	Moderate	Highly Likely	A clear project manager should be appointed to the development team. Set up a project charter wherein all team roles, expectations, conflict resolution processes, communication, and the like are clearly defined. All team members should sign off after being given time to understand and potentially negotiate the terms. Regular team and individual check-ups should be conducted.
14	Lack of resources: allocated resources are below the amount needed to complete the project.	Moderate	Possible	During planning, conduct resource requirement analysis and implement contingency plans for resource procurement.
15	Budget constraints: available funds are below the amount needed to complete the project.	Moderate	Possible	Construct a detailed budget with contingency reserves. Monitor expenses and conduct regular budget reviews.
16	Missed deadlines: Tasks are not completed by the associated deadline.	Moderate	Possible	Construct a detailed project schedule, accounting for all deadlines and slight buffer times to critical tasks. Utilise a network diagram or a Gantt chart.
17	Weak sponsor engagement: Difficulty communicating, acquiring assistance, etc.	Moderate	Possible	Establish communication channels for emergency escalation, such as alternative means of contact.
18	Legislation changes: changes in relevant legislation (i.e. POPIA, Children's Act, Child Justice Act) will require unplanned changes in the project to ensure compliance	Major	Highly Unlikely	Conduct regular research and analysis on relevant legislation to ensure continued compliance. Design the system with configurable business rules to ensure quick legal compliance capability

Table 8: Risk Matrix